CERTIFICATE OF INSTALLATION

Note: This table completed by ~~HERS~~ECC Registry.

Project Name:	Enforcement Agency:
Dwelling Address:	Permit Number:
City and Zip Code:	Permit Application Date:

A. General Information

Notes:

- The outdoor design temperatures for heating shall be ≥99.0% Heating Dry Bulb or the Heating Winter Median of Extremes values.
- The outdoor design temperatures for cooling shall be ≤1.0% Cooling Dry Bulb and Mean Coincident Wet Bulb values.
- In fields A13 and A14,
    1. If the airflow is verified to be 350 CFM/ton or higher there is no maximum capacity (Section 150.2(a)1Eii) and user should enter N/A.
    2. If the space conditioning system is ductless (Exception 1 to Section 150.2(a)1Eii), user should enter N/A.

01	Dwelling Unit Name		02	Climate Zone	
03	Dwelling Unit Total Conditioned Floor Area (ft²)		04	Number of Space Conditioning Systems in this Dwelling Unit	
05	Certificate of Compliance Type		06	Method Used to Calculate HVAC Loads (See Section 150.0(h))	
07	Outdoor Design Condition Source (See Section 150.0(h)2		08	Cooling Outdoor Design Temperature Selected (°F)	
09 ~~7~~	~~Calculated Dwelling Unit Sensible Cooling Load (Btu/h)~~ Heating Outdoor Design Temperature Selected (°F)		10 ~~08~~	~~Calculated Dwelling Unit Heating Load (Btu/h)~~ Calculated Dwelling Unit Sensible Cooling Load (Btu/h	
11 ~~09~~	~~Dwelling Unit Number of Bedrooms~~ Calculated Dwelling Unit Heating Load (Btu/h)		12	Dwelling Unit Number of Bedrooms	
This Table reports for Addition only, if project scop on the CF1R do not have Addition, then display the “Fields A13–A16 do not apply” message					
13	Maximum Heating Capacity According to Table 150.2-A (Btu/h)		14	Maximum Cooling Capacity According to Table 150.2-B (Btu/h)	
15	Envelope Leakage Specified in Load Calculation (ACH)		16	Source of Envelope Leakage Specified in Load Calculation	

MCH-01b - Space Conditioning Systems Ducts and Fans - Prescriptive Alterations

B. Space Conditioning (SC) System Information

01	02	03	04	05	06	07	08	09	10
SC System ID/Name from CF1R	SC System Description of Area Served	CFA served by this SC System (ft^2):	Is the SC system a ducted system?	Does work include installing a refrigerant containing component?	Does work include installing new SC System components?	Does work include installing more than 25 feet of ducts?	Does work include installing entirely new duct system?	Does work include installing entirely new SC system?	Alteration Type
Notes:									

C. Space Conditioning (SC) System Alterations Compliance Information

01	02	03	04	05	06	07	08	09	10	10b	11	12	13	14
SC System ID/Name from CF1R	SC System Description of Area Served	Heating System Type	Altered Heating Component	Heating Efficiency Type	Heating Minimum Efficiency Value	Cooling System Type	Altered Cooling Components	Cooling Efficiency Type	Cooling Minimum Efficiency Value SEER/SEER2	Cooling Minimum Efficiency Value EER/EER2 /CEER	Required Thermostat Type	Number of Indoor Units for this System	Number of Ducted Indoor Units for this System	Central Fan Integrated (CFI) Ventilation System Status
Notes:														

D. Installed Heating Equipment Information for Gas Furnace Indoor Unit, or Heat Pump Indoor Unit, or Packaged Unit (Gas Furnace or Heat Pump)

01	02	03	04	05	06	07	08	09	10
								Multi-Split Systems only	
SC System ID/Name from CF1R	SC System Description of Area Served	Heating Efficiency Type	Heating Efficiency Value	Heating Unit Manufacturer	Heating Unit Model Number	Heating Unit Serial Number	Rated Heating Capacity, Output (Btu/h)	Indoor Unit Name or Description of Area Served	Indoor Unit Duct Status
Notes:									

E. Installed Cooling Equipment Information for Outdoor Condenser or Package Unit (Air Conditioner or Heat Pump)

01	02	03	04	04b	05	06	07	08	09
SC System ID/Name from CF1R	SC System Description of Area Served	Cooling Efficiency Type	Cooling Efficiency Value SEER/SEER2	Cooling Efficiency Value EER/EER2/CEER	Condenser or Package Unit Manufacturer	Condenser or Package Unit Model Number	Condenser or Package Unit Serial Number	System Cooling Capacity at Design Conditions (Btu/h)	Condenser Nominal Capacity (tons)
Notes:									

F. Altered Space Conditioning System Duct Information (<75% of duct system is altered; or duct system is not altered)

01	02	03	04	05	06	07	08	09	10	11	12
SC System ID/Name from CF1R	SC System Description of Area Served	Indoor Unit Name or Description of Area Served	Was Any New Ducting Installed?	Required New Duct R-Value	Installed New Supply Duct Location	Installed New Supply Duct R-Value	Installed New Return Duct Location	Installed New Return Duct R-Value	Exception from Min R-Value	Can Approved Airflow Protocols be used to test this System?	Indoor Unit Nominal Cooling Capacity (tons)
Notes:											

G. Installed New or Complete Replacement Duct System information

01	02	03	04	05	06	07	08	09	10	11	12	13	14	15
SC System ID/Name from CF1R	SC System Description of Area Served	Indoor Unit Name or Description of Area Served	Indoor Unit Total Duct Length	Required New Duct R-Value (Unconditioned Space)	Supply Duct Location	New or Replaced Supply Duct R-Value	Return Duct Location	New or Replaced Return Duct R-Value	Exception from Min R-Value	Method of Compliance with Airflow and Fan Efficacy Req's in 150.0(m)13	Number of Air Filter Devices on Indoor Unit	Can Approved Airflow Protocols be used to test this System?	Can Approved Fan Efficacy Protocol be used to test this System?	Indoor Unit Nominal Cooling Capacity (tons)
Notes:														

H. Installed Air Filter Device Information

Mandatory requirements for air filter devices are specified Section 150.0(m)12. The installer shall place a sticker in or near each filter grille that displays the design airflow rate for that filter grille/rack and the maximum allowed clean filter pressure drop at the design airflow rate. This will inform the occupant of the airflow vs pressure drop performance required for replacement air filters.

01	02	03	04	05	06	07	08	09	10	11	12	13
SC System ID/Name from CF1R	SC System Description of Area Served	Indoor Unit Name or Description of Area Served	Air Filter Name or Description of Location	Air Filter Rack Type	Design Airflow Rate for Air Filter Device (cfm)	Air Filter Nominal Depth (inch)	Air Filter Nominal Length (inch)	Air Filter Nominal Width (inch)	Air Filter Calculated Nominal Face Area (inch^2)	Air Filter Required Minimum Face Area (inch^2)	Face Area Compliance	Design Allowable Pressure Drop for Air Filter Device (inch W.C.)
Notes:												

I. Air Filter Device Requirements

Mandatory Air Filter Device Requirements can be found in Section 150.0(m)12A-E. Some mandatory requirements may apply in addition to those listed below.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	All recirculated air and all outdoor air (including make up air) supplied to the occupiable space is filtered before passing through the system's thermal conditioning components.
02	The space conditioning system shall be designed to accommodate the clean-filter pressure drop imposed by the system air filter device(s). The design airflow rate and maximum allowable clean-filter pressure drop at the design airflow rate applicable to each air filter shall be determined by the system designer. The system installer shall affix a sticker/label to each system air filter grille/rack location that discloses the filter's design airflow rate and the filter's maximum allowable clean-filter pressure drop at the design airflow rate. The sticker/label shall be permanently affixed to the air filter grille/rack, readily legible, and visible to a person replacing the air filter.
03	All system air filter devices shall be located and installed in such a manner as to allow access and regular service by the system owner.
04	The system shall be provided with air filters having a designated efficiency equal to or greater than MERV 13 when tested in accordance with ASHRAE Standard 52.2, or a particle size efficiency rating equal to or greater than 50 percent in the 0.30-1.0 µm range and equal to or greater than 85 percent in the 1.0-3.0 µm range when tested in accordance with AHRI Standard 680.
05	The system shall be provided with air filters that have been labeled by the manufacturer to disclose efficiency and pressure drop ratings that conform to the efficiency and pressure drop requirements for the air filter grilles/racks.
06	Filter racks or grilles shall use gaskets, sealing, or other means to close gaps around inserted filters and prevent air from bypassing the filter.

J. ~~HERS~~ECC Verification Requirements for Duct Systems

01	02	03	04	05	06	07	08	09
SC System Identification or Name	SC System Description of Area Served	Indoor Unit Name or Description of Area Served	Exemption From Duct Leakage Requirements	MCH-20 Duct Leakage Test	MCH-21 Duct Location Verification	MCH-22 AHU Fan Efficacy (W/cfm)	MCH-23 AHU Airflow Rate (cfm/ton)	MCH-28 Return Duct Design - Table 150.0-B or C
Notes:								

K. ~~HERS~~ECC Verification Requirements for Space Conditioning Equipment

01	02	03
SC System ID/Name from CF1R	SC System Description of Area Served	MCH-25 Refrigerant Charge
Notes:		

L. Space Conditioning Systems, Ducts and Fans – Mandatory Requirements and Additional Measures

Additional mandatory requirements from Section 150.0 that are not listed here may be applicable to some systems. These requirements may be applicable to only newly installed equipment or portions of the system that are altered. Existing equipment may be exempt from these requirements.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

Heating Equipment

01	Equipment Efficiency: All heating equipment must meet the minimum efficiency requirements of Section 110.1 and Section 110.2(a) and the Appliance Efficiency Regulations.
02	Controls: All unitary heating systems, including heat pumps, must be controlled by a setback thermostat. These thermostats must be capable of allowing the occupant to program the temperature set points for at least four different periods in 24 hours. See Sections 150.0(i), 110.2(c).
03	Sizing: Heating load calculations must be done on portions of the building served by new heating systems to prevent inadvertent undersizing or oversizing. See sections 150.0(h)1 and 2~~)~~, and the exceptions.
04	Furnace Temperature Rise: Central forced-air heating furnace installations must be configured to operate at or below the furnace manufacturer's maximum inlet-to-outlet temperature rise specification. See Section 150.0(h)4.
05	Standby Losses and Pilot Lights: Fan-type central furnaces may not have a continuously burning pilot light. Section 110.5 and Section 110.2(d).

Cooling Equipment

06	Equipment Efficiency: All cooling equipment must meet the minimum efficiency requirements of Section 110.1 and Section 110.2(a) and the Appliance Efficiency Regulations.
07	Refrigerant Line Insulation: All refrigerant line insulation in split system air conditioners and heat pumps must meet the R-value and protection requirements of Section 150.0(j)1~~2~~ and 2~~3~~, and Section 150.0(m)9.
08	Condensing Unit Location: Condensing units shall not be placed within 5 feet of a dryer vent outlet. See Section 150.0(h)3A.
09	Liquid Line Filter Drier: A liquid line filter drier shall be installed according to the manufacturer's specifications 150.0(h)3B.
10	Sizing: Cooling load calculations must be done on portions of the building served by new cooling systems to prevent inadvertent undersizing or oversizing. See Section 150.0(h)1 and 2, and the exception.

Cooling and Heating Equipment (Additional Requirement)

11	**System Selection:** See section 150.0(h)5 A. Equipment sizing and selection shall meet the cooling and heating loads of Section 150.0(h)1 and 2. B. Systems shall be sized based on ACCA Manual S-2023 in accordance with these requirements: i. **Cooling Capacity:** There is no limit on the minimum capacity. ii. **Furnaces**: Heating capacity shall be sized based on ACCA Manual S-2023, Table N2.5. iii. **Heat Pump Heating Capacity**: a. Minimum: Heating systems are required to have a heating capacity meeting the minimum requirements of the CBC not including any supplementary heating. b. Maximum: There is no limit on the maximum heating capacity.
12	**Defrost:** See section 150.0(h)6 and the exceptions. A. If a heat pump is equipped with an installer adjustable defrost delay timer, the delay timer shall be set to greater than or equal to 90 minutes. B. The installer shall certify on the Certificate of Installation (CF2R) that the control configuration has been tested in accordance with the testing procedure in the CF2R. **Exception 1 to Section 150.0(h)6.** Dwelling units in Climate Zones 6 and 7. **Exception 2 to Section 150.0(h)6.** Dwelling units with a conditioned floor area of 500 square feet or less in Climate Zones 3, 5 through 10, and 15.

13	**Supplementary heating control configuration:** See section 150.0(h)7. Heat pumps with supplementary heat, including, but not limited to, electric resistance heaters or gas furnace supplementary heating, shall comply with the following requirements: **See section 150.0(h)7 for exceptions.** A. Lock out supplementary heating above an outdoor air temperature of no greater than 35°F. There are additional thermostat requirements in section 150.0(i)2. B. The installer shall certify on the Certificate of Installation that the control configuration has been tested in accordance with the testing procedure found in the CF2R. C. The controls may allow supplementary heater operation above 35°F only during defrost; or when the user selects emergency operation.
14	**Sizing of electric resistance supplementary heat:** See section 150.0(h)8. For heat pumps with electric resistance heating, the capacity of electric resistance heat shall not exceed the heat pump nominal cooling capacity (at 95°F ambient conditions) multiplied by 2.7 kW per ton, rounded up to the closest kW.
15	**Capacity variation with third-party thermostats:** See section 150.0(h)9 Variable or multi-speed systems shall comply with the following requirements: A. The space conditioning system and thermostat together shall be capable of responding to heating and cooling loads by modulating system compressor speed, and meet thermostat requirements in section 150.0(i)2.

Air Distribution System Ducts, Plenums and Fans

~~11~~ 16	Insulation: The minimum duct insulation value is R-6 or ducts can be uninsulated if the duct system is located entirely in conditioned space. Note that higher values may be required by the prescriptive or performance requirements. See Section 150.0(m)1B for exceptions.
~~12~~ 17	Connections and Closures: All installed air-distribution system ducts and plenums must meet the requirements of CMC Sections 601.0, 602.0, 603.0, 604.0, 605.0 and ANSI/SMACNA-006-2006.

Heat Pump Thermostat

~~13~~ 18	A thermostat shall be installed that meets the requirements of Section 110.2(b) and Section 110.2(c).
~~14~~ 19	The thermostat shall be installed in accordance with the manufacturers published installation specifications.
~~15~~ 20	First stage of heating shall be assigned to heat pump heating.
~~16~~ 21	Second stage back up heating shall be set to come on only when the indoor set temperature cannot be met.
22	Setback thermostats: All heating or cooling systems, including heat pumps, not controlled by a central energy management control system (EMCS) shall have a setback thermostat, as specified in Section 110.2(c). See Section 150.0(i)1
23	Thermostats that are applied to heat pumps with supplemental heating: See Section 150.0(i)2 The thermostats controlling heat pumps with electric resistance supplementary heat or gas furnace supplementary heat shall comply with the following requirements: See Section 150.0(i)2 for exceptions. A. The thermostat shall receive outdoor air temperature from an outdoor air temperature sensor or from an internet weather service. B. B. The thermostat shall display the outdoor air temperature. C. The thermostat and heat pump shall lock out supplementary heat when the outdoor air temperature is above 35°F. D. The thermostat shall have an indicator to notify when supplementary heat or emergency heat is in use. E. During defrost or when the user selects emergency heating, supplementary heat operation is permitted above 35°F.

Space Conditioning Load Calculations and System Capacity for Additions

24	Minimum capacity limits and supplemental heating requirements are as described in Section 150.0(h)
25	The maximum capacity depends on the relative sizes of the calculated heating design load (HL) and cooling design load (CL), the type of space conditioning system, and the duct sizing. **See section 150.2(a)1Eii for exceptions** a. In situations where airflow is field verified to be at least 350 cfm/ton, there is no maximum capacity limit. b. In situations where airflow is NOT field verified to be at least 350 cfm/ton, the system capacities shall be no larger than indicated in Table 150.2-A for heating and Table 150.2-B for cooling.
26	For additions, the envelope leakage specified in the load calculation shall be no greater than the values shown in Table 150.2-C. **See section 150.2(a)1Eiii for exceptions**

M. Test of Defrost Delay Timer Setting

The installing contractor shall confirm that a heat pump’s installer-adjustable Defrost Delay Timer Setting (if it exists) is set to no less than 90 minutes.

01	Test Applicability. Select the statement describing test applicability for this project: 1. The test applies because the heat pump utilizes an installer adjustable Defrost Delay Timer Setting to control defrost and there are no exceptions. 2. The test does not apply because the heat pump does not utilize an installer-adjustable Defrost Delay Timer Setting to control defrost. 3. The test does not apply because Exception 1. Dwelling units in Climate Zones 6 and 7 applies. 4. The test does not apply because Exception 2 . Dwelling units with a conditioned floor area of 500 square feet or less in Climate Zones 3, 5 through 10, and 15 applies.	
02	Recording Configuration of Controls. Specify the mechanism for setting the Defrost Delay Timer Setting (for example, the name of defrost delay timer setting in the thermostat setup, or the location and number of the specific dip switch, jumper, or dial that adjusts Defrost Delay timer).	
03	Record the heat pump’s Maximum Available Defrost Delay Timer Setting (minutes).	
04	Record where you set the Defrost Delay Timer Setting (fo example, the numeric timer setting, dip switch position, jumper configuration, or dial setting).	
05	Record where you set the Defrost Delay Timer Setting, in minutes.	
06	Confirming Configuration of Controls. If possible, the Defrost Delay Timer Setting must be 90 minutes or greater. Confirm the Defrost Delay Timer Setting is at least 90 minutes.	

N. Test of Supplementary Heating Lockout

The installing contractor shall confirm the heat pump or thermostat is configured to lock out supplementary heating anytime the outdoor air temperature is above 35°F.

01	Test Applicability. Select the statement describing test applicability for this project: 1. The test applies because the heating system is a heat pump with supplementary heating and there are no exceptions. 2. The test does not apply because the heating system does not include a heat pump with auxiliary heating. 3. The test does not apply because Exception 1. Heat pump being installed in a single family dwelling unit in Climate Zones 7 or 15. 5. The test does not apply because Exception 2 . Heat pump being installed in a single family dwelling unit with a conditioned floor area of 500 square feet or less.	
02	If supplementary heating is electric resistance, what is its rated capacity (kW)?	
03	If supplementary heating is gas, what is its rated input capacity (kBtu/hr)?	
04	Confirming Configuration of Controls. Specify the mechanism for locking out the supplementary heating (for example, the name of supplementary heating lockout setting in the thermostat setup, or the location and number of the specific dip switch, jumper, or dial that adjusts lockout temperature). If there is no mechanism to lock out supplementary heating above a temperature no greater than 35°F this test fails.	
05	Record supplementary heating lock-out setting (for example, the numeric thermostat lock-out setpoint, dip switch position, jumper configuration, or dial setting).	
06	At what Outdoor Air Temperature (OAT) are the controls configured to begin locking out supplementary heating? If this number is above 35°F this test fails.	

DOCUMENTATION AUTHOR'S DECLARATION STATEMENT

1. I certify that this Certificate of Installation documentation is accurate and complete

Documentation Author Name:	Documentation Author Signature:
Documentation Author Company Name:	Date Signed:
Address:	CEA/~~HERS~~ECC Certification Identification (If applicable):
City/State/Zip:	Phone:

RESPONSIBLE PERSON'S DECLARATION STATEMENT

I certify the following under penalty of perjury, under the laws of the State of California:

1. The information provided on this Certificate of Installation is true and correct.
2. I am either: a) a responsible person eligible under Division 3 of the Business and Professions Code in the applicable classification to accept responsibility for the system design, construction, or installation of features, materials, components, or manufactured devices for the scope of work identified on this Certificate of Installation, and attest to the declarations in this statement, or b) I am an authorized representative of the responsible person and attest to the declarations in this statement on the responsible person's behalf.
3. The constructed or installed features, materials, components or manufactured devices (the installation) identified on this Certificate of Installation conforms to all applicable codes and regulations and the installation conforms to the requirements given on the Certificate of Compliance, plans, and specifications approved by the enforcement agency.
4. I understand that a registered copy of this Certificate of Installation shall be posted or made available with the building permit(s) issued for the building and shall be made available to the enforcement agency for all applicable inspections. I will take the necessary steps to fulfill~~ensure~~ this requirement~~ is accomplished~~.
5. I understand that a registered copy of this Certificate of Installation is required to be included with the documentation the builder provides to the building owner at occupancy. I will take the necessary steps to fulfill~~ensure~~ this requirement~~ is accomplished~~.

Responsible Builder/Installer Name:	Responsible Builder/Installer Signature:	
Company Name: (Installing Subcontractor or General Contractor or Builder/Owner)	Position With Company (Title):	
Address:	CSLB License:	
City/State/Zip:	Phone:	Date Signed:

CF2R-MCH-01b-E User Instructions

Minimum requirements for prescriptive HVAC installation compliance can be found in Building Energy Efficiency Standards Section 150.2(b)1C.

Completing these documents will require that you have the Reference Appendices for the ~~2022~~2025 Building Energy Efficiency Standards. This document contains the Joint Appendices which are used to determine climate zone and to complete the section for opaque surfaces.
When the term CF2R is used it means the CF2R-MCH-01-H.

Instructions for sections with column numbers and row numbers are given separately.

Section A. General Information

1. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document.
2. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document.
3. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document. When the project scope includes an addition to an existing building, the value is equal to the sum of the existing conditioned floor area plus the conditioned floor area of the addition. The default value from the CF1R may be overwritten in this document. Overwriting the default value will automatically flag this entry and subject it to additional scrutiny by QA and enforcement personnel.
4. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document. This value may be overwritten in this document but valid discrepancies with the CF1R are uncommon. Overwriting the default value will automatically flag this entry and subject it to additional scrutiny by QA and enforcement personnel.
5. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document.
6. Oversized equipment can result in reduced efficiency and capacity. Entirely new systems (see definition in Section 9.6.9 of the RCM) must be properly sized to match the heating and cooling load of the space that it serves. To do this, heating and cooling load calculations must be performed using an approved calculation methodology. These are listed here. Select the load calculation methodology used for this dwelling unit. If the project consists of a partial replacement of equipment or ducts (change-out) then load calculations are not required. Select N/A. Load calculations are always recommended, especially if the loads of the house have been changed since the original equipment has been installed (reduced via weatherization, other improvements).
7. Enter the Outdoor Design Condition Source (See Section 150.0(h)2), user select from the list.
8. Enter the Cooling Outdoor Design Temperature Selected (°F) for the dwelling unit described by this document.
9. Enter the Heating Outdoor Design Temperature Selected (°F) for the dwelling unit described by this document.

~~7.~~10. Enter the total sensible cooling load for the dwelling unit described by this document. For projects involving dwelling units with more than one system, this will be a sum of the loads for the parts of the dwelling unit served by those systems. If the project consists of a partial replacement of equipment or ducts (change-out), then load calculations are not required. Select N/A.

~~8.~~11. Enter the total heating load for the dwelling unit described by this document. For projects involving dwelling units with more than one system, this will be a sum of the loads for the parts of the dwelling unit served by those systems. If the project consists of a partial replacement of equipment or ducts (change-out), then load calculations are not required. Select N/A.

12. Enter the number of bedrooms in the dwelling unit.

13. User select from the list or user input. If airflow is field verified to be 350 CFM/ton or higher, or if the space conditioning system is ductless, then enter N/A.

14. User input. If airflow is field verified to be 350 CFM/ton or higher, or if the space conditioning system is ductless, then enter N/A.

15. User input. Referenced from Table 152.2-C and Field Verification and Diagnostic Testing.

~~9.~~16. User select from the list.

Section B. Space Conditioning (SC) System Information

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10. This field is filled out automatically based on the entries in the previous columns.

Section C. Space Conditioning (SC) System Alterations Compliance Information

1. This field is filled out automatically. It is referenced from the previous section.
2. This field is filled out automatically. It is referenced from the previous section.
3. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document. This value may be overwritten in this document but valid discrepancies with the CF1R are uncommon. Overwriting the default value will automatically flag this entry and subject it to additional scrutiny by QA and enforcement personnel. Revising the CF1R to match is recommended and may be required.
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7. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document. This value may be overwritten in this document but valid discrepancies with the CF1R are uncommon. Overwriting the default value will automatically flag this entry and subject it to additional scrutiny by QA and enforcement personnel. Revising the CF1R to match is recommended and may be required.
8. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document. This value may be overwritten in this document but valid discrepancies with the CF1R are uncommon. Overwriting the default value will automatically flag this entry and subject it to additional scrutiny by QA and enforcement personnel. Revising the CF1R to match is recommended and may be required.
9. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document. This value may be overwritten in this document but valid discrepancies with the CF1R are uncommon. Overwriting the default value will automatically flag this entry and subject it to additional scrutiny by QA and enforcement personnel. Revising the CF1R to match is recommended and may be required.
10. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document. This value may be overwritten in this document but valid discrepancies with the CF1R are uncommon. Overwriting the default value will automatically flag this entry and subject it to additional scrutiny by QA and enforcement personnel. Revising the CF1R to match is recommended and may be required.

10b. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document. This value may be overwritten in this document but valid discrepancies with the CF1R are uncommon. Overwriting the default value will automatically flag this entry and subject it to additional scrutiny by QA and enforcement personnel. Revising the CF1R to match is recommended and may be required

11. This field is filled out automatically. It is calculated based on entries in previous columns.
12. If the space conditioning system is a multiple-split system, then enter the total number of indoor units (ducted and ductless) connected to the outdoor unit. If the system is a type that does not have an outdoor unit, such as a heating-only type that uses only a furnace air-handling unit, enter 1 for the number of indoor units (The furnace air-handling unit is an indoor unit).
13. If the space conditioning system is a multiple-split system, then enter the number of ducted indoor units (AHU) connected to the outdoor unit. If the system is a type that does not have an outdoor unit, such as a heating-only type that uses only a furnace air-handling unit, enter 1 for the number of indoor units (The furnace air-handling unit is an indoor unit).

14. If the indoor unit is used to bring outdoor air into the dwelling, the system may be used to comply with the IAQ mechanical ventilation requirements. This is called central fan integrated ventilation (CFI). Select CFI System if the system is used to provide IAQ ventilation.

Section D. Installed Heating Equipment Information

1. This field is filled out automatically. It is referenced from a previous section.
2. This field is filled out automatically. It is referenced from a previous section.
3. This field is filled out automatically. It is referenced from a previous section
4. Enter the certified heating efficiency of the *installed* equipment. This value is verified against the minimum value shown in Section C. The installed efficiency must be greater than or equal to the required minimum efficiency.
5. Enter the name of the *installed* Heating Unit Manufacturer as shown on the equipment nameplate.
6. Enter the name of the *installed* Heating Unit Model Number as shown on the equipment nameplate.
7. Enter the name of the *installed* Heating Unit Serial number as shown on the equipment nameplate.
8. Enter the rated heating capacity (output) of the *installed* Heating Unit in BTUs per hour.
9. Enter text to provide a name for multi-split indoor units if prompted to do so, otherwise the field is filled out automatically.
10. Select the description that best describes the distribution system if prompted to do so (allowed values are 1:[Ductless] 2:[Ducted >10ft length] 3:[Ducted ≤10ft length], otherwise the field is filled out automatically.

Section E. Installed Cooling Equipment Information:

1. This field is filled out automatically. It is referenced from a previous section.
2. This field is filled out automatically. It is referenced from a previous section.
3. This field is filled out automatically. It is referenced from Section C.
4. Enter the certified cooling efficiency of the *installed* equipment that corresponds to the type shown in the previous column. This value is verified against the minimum value shown in Section C. The installed efficiency must be greater than or equal to the required minimum efficiency.

4b. Enter the certified cooling efficiency of the installed equipment that corresponds to the type shown in the previous column. This value is verified against the
minimum value shown in Section C. The installed efficiency must be greater than or equal to the required minimum efficiency.

4. Enter the name of the *installed* Condenser or Package Unit Manufacturer as shown on the equipment nameplate.
5. Enter the name of the *installed* Condenser or Package Unit Model Number as shown on the equipment nameplate.
6. Enter the name of the *installed* Condenser or Package Unit Serial Number as shown on the equipment nameplate.
7. Enter the sensible cooling capacity at design conditions of the *installed* cooling system in BTUs per hour.
8. Enter the *installed* Condenser Nominal Cooling Capacity in tons. Note that this is based on the condenser, not the coil or air handler. This can usually be determined by the condenser model number.

Section F. Extension of Existing Duct System, Greater Than 25 Feet

1. This field is filled out automatically. It is referenced from a previous section.
2. This field is filled out automatically. It is referenced from a previous section.
3. Enter a brief name or description of the indoor unit area served. Examples: Master Bedroom, Dining Room, Living Room, etc.
4. If any lengths of new ducts were installed, answer yes, otherwise if new ducts were not installed, answer no.
5. This field is filled out automatically based on values referenced from other sections.
6. Select the choice that best describes the predominant location of the supply ducts for this system
7. Enter the R-value of the *installed* supply ducts. This value is verified against the minimum value required by the standards. The installed R-value must be greater than or equal to the required minimum R-value.
8. Select the choice that best describes the predominant location of the return ducts for this system
9. Enter the R-value of the installed return ducts. This value is verified against the minimum value required by the standards. The installed R-value must be greater than or equal to the required minimum R-value
10. The duct system needs to meet minimum R-6 requirement except for portions of ducts located in conditioned space. Duct systems that are entirely in conditioned space can be uninsulated, subject to ~~HERS~~ECC verification. If the system is of a type that can use one of the approved protocols for testing the airflow rate, then enter yes. Otherwise enter no. Most ducted split systems and package systems are of the type that minimum airflow can be verified using an approved measurement procedure. Examples of systems that do not meet this description are ductless systems. A "No" response here may subject the project to additional scrutiny by enforcement personnel. Note: that the protocol in RA3.3.3.1.5 (Alternative to Compliance with Minimum System Airflow Requirements for Altered Systems) is not one of the protocols that is allowed to be used to justify a "yes" to this question.
11. Enter the indoor unit nominal cooling capacity (tons) if the indoor unit is a multiple-split system type, otherwise this field is not needed.

Section G. Installed Duct System information

1. This field is filled out automatically. It is referenced from the same row and column in the previous sections.
2. This field is filled out automatically. It is referenced from the same row and column in the previous sections.
3. Enter a brief name or description of the indoor unit area served. Examples: Master Bedroom, Dining Room, Living Room, etc..
4. Enter the description of the total combined length of the supply and return ducts on this indoor unit. The possible choices are: >10ft length, and ≤10ft length.
5. This field is filled out automatically. This is the minimum R-value for new ducts in this climate zone.
6. Select the choice that best describes the predominant location of the supply ducts for this system.
7. Enter the R-value of the *installed* supply ducts. This value is verified against the minimum value in G05. The installed R-value must be greater than or equal to the minimum R-value.
8. Select the choice that best describes the predominant location of the return ducts for this system.

9. Enter the R-value of the *installed* return ducts. This value is verified against the minimum value shown in Section C. The installed R-value must be greater than or equal to the required minimum R-value.
10. The duct system needs to meet minimum R-6 requirement except for portions of ducts located in conditioned space. Duct systems that are entirely in conditioned space can be uninsulated, subject to ~~HERS~~ECC verification.
11. Pick the appropriate choice. Refer to section 150.0(m)13 of the ~~2022~~2025 Building Energy Efficiency Standards, and Section 4.4 of Chapter 4 of the ~~2022~~2025 Residential Compliance Manual for more information.
12. Specify the number of air filter devices installed on this indoor unit. Air filter devices installed in completely new systems must be properly sized, as documented in the next section. The value entered here will determine the number of rows needed in the following section.
13. If the system is of a type that can use one of the approved protocols for testing the airflow rate, then enter yes. Otherwise enter no. Most ducted split systems and package systems are of the type that minimum airflow can be verified using an approved measurement procedure. Examples of systems that do not meet this description are ductless systems. A "No" response here may subject the project to additional scrutiny by enforcement personnel. Note: that the protocol in RA3.3.3.1.5 (Alternative to Compliance with Minimum System Airflow Requirements for Altered Systems) is not one of the protocols that is allowed to be used to justify a "yes" to this question.
14. If the system is of a type that can use one of the approved protocols for testing the fan efficacy, then enter yes. Otherwise enter no.
15. Enter the indoor unit cooling capacity if the indoor unit is a multiple-split system type, otherwise this field is not needed.

Section H. Installed Air Filter Device Information

1. This field is filled out automatically. It is referenced from the same row and column in the previous sections.
2. This field is filled out automatically. It is referenced from the same row and column in the previous sections.
3. This field is filled out automatically. It is referenced from the same row and column in the previous sections.
4. Enter a descriptive name of each air filter device so that it may be distinguished from others in the same system. Examples: FG1, filter2, etc.
5. Select the appropriate type of filter device from the list.
6. Enter the design flow in CFM of the filter device. The total for all filter devices in a single system should be greater than or equal to the total system design CFM in cooling mode (or heating mode for heat-only systems).
7. Enter the nominal depth of the filter in inches. This is the dimension that is parallel to the airflow. many filters available for sale are 1-inch depth. The ~~2022~~2025 standards encourage use of 2-inch depth filters.
8. Enter the nominal length of the filter. for example, if the filter is 20" x 30", enter 30.
9. Enter the nominal width of the filter, for example, if the filter is a 20" x 30", enter 20.
10. This field is calculated automatically based on your entries in 8 and 9.
11. This value is calculated automatically for 1-inch depth filters. 2-inch depth or greater filters may use a value determined by the system designer.

12. This field determines whether a 1-inch depth filter complies with the sizing requirements in section 150.0(m)12. A 2-inch depth or greater filter may use the face area determined by the system designer, however most systems have to meet airflow rate and fan efficacy requirements.
13. Enter the design static pressure drop determined by the system designer if 2-inch or greater filters are used. For 1-inch depth filters, the maximum pressure drop is mandatory 0.1 inch W.C.. Filters installed in the filter grille/rack must be capable of meeting this maximum pressure drop at the design airflow rate, as shown on the manufacturer's filter label. Not accounting for higher filter pressure drops will result in poor system airflow characteristics, reduced capacity and reduced efficiency. This may result in not passing field verification.

Section I. Air Filter Device Requirements

This table is a list of requirements for air filter devices.

Section J. ~~HERS~~ECC Verification Requirements

1. This field is filled out automatically. It references previous sections in this document.
2. This field is filled out automatically. It references previous sections in this document.
3. This field is filled out automatically. It references previous sections in this document.
4. If applicable, select from the available exemptions listed. Exemptions will be flagged and may subject the system to additional enforcement scrutiny.
5. This field is filled out automatically. It is calculated based on data from the CF1R and from previous sections in this document.
6. This field is filled out automatically. It is calculated based on data from the CF1R and from previous sections in this document.
7. This field is filled out automatically. It is calculated based on data from the CF1R and from previous sections in this document.
8. This field is filled out automatically. It is calculated based on data from the CF1R and from previous sections in this document.
9. This field is filled out automatically. It is calculated based on data from the CF1R and from previous sections in this document.

Section K. ~~HERS~~ECC Verification Requirements for Space Conditioning Equipment

1. This field is filled out automatically. It is calculated based on data from the CF1R and from previous sections in this document.
2. This field is filled out automatically. It is calculated based on data from the CF1R and from previous sections in this document.
3. This field is filled out automatically. It is calculated based on data from the CF1R and from previous sections in this document.

Section L. Space Conditioning Systems, Ducts and Fans – Mandatory Requirements and Additional Measures

This table is a list of mandatory measures and additional requirements for space conditioning systems, ducts and fans.

Section M. Test of Defrost Delay Timer Setting (Section 150.0(h)6)
This table is certification requirements for Test of Defrost Delay Timer Setting

Section N. Test of Supplementary Heating Lockout (Section 150.0(H)7
This table is certification requirements for Test of Supplementary Heating Lockout

A. General Information					
01	Dwelling Unit Name	<<default reference text from CF1R; or allow user override input: text, 15 character maximum>>	02	Climate Zone	<<default reference text from CF1R >>
03	Dwelling Unit Total Conditioned Floor Area (ft^2)	<<numeric: xxxxx; **if1 parent is CF1R-PRF**, then if2 project scope = Newly Constructed (Addition Alone) then prompt user to enter a value equal to dwelling unit existing CFA + addition CFA else reference the value from CF1R endif2 **elseif parent is CF1R-NCB-01**, then **if3 project scope = New Addition greater than 1,000 ft2** then prompt user to enter a value equal to dwelling unit existing CFA + addition CFA **elseif project scope = Newly Constructed Building, then** if4 building type = Single Family, then reference value from CF1R-NCB field A10 **elseif parent is CF1R-ADD-01**, then if5 building type= Single Family, then reference value from field A08 from the CF1R-ALT-02 that is required for the dwelling unit according to CF1R-ADD-01 Section J. **elseif parent is CF1R-ALT-01**, then if6 building type= Single Family, then reference value from field A08 from the CF1R-ALT-02 that is required for the dwelling unit according to CF1R-ALT-01 Section G. elseif parent is CF1R-ALT-02, then reference value from CF1R-ALT-02 field A08. endif1 allow user to override default and input a value; flag overridden values and report in project status notes field >>	04	Number of Space Conditioning Systems in this Dwelling Unit	<<integer: xx; If parent is CF1R-ALT-02 doc type, then use as default the value referenced from CF1R-ALT-02 Section A (field A10); or allow user to override the default and input a new value; flag non-default values and report in project status notes field; elseif parent is not CF1R-ALT-02 doc type, then user input the integer value>>

05	Certificate of Compliance Type	<< reference document type property from CF1R: allowed values: performance (CF1R-PRF); or prescriptive additions/alterations (CF1R-ADD/CF1R-ALT); or prescriptive newly constructed (CF1R-NCB)>>	06	Method Used to calculate HVAC loads (See Section 150.0(h).)	<<user select from list: *ASHRAE Handbook; *SMACNA Residential Comfort System Installation Standards Manual; *ACCA Manual J * n/a equipment changeout, like-for-like *Exception 1 to Section 150.0(h)1: Block loads, (the total load for all rooms combined that are served by the central equipment,) may be used for the purpose of system sizing for additions >>
07	Outdoor Design Condition Source (See Section 150.0(h)2	<< user select all that apply: *Reference Joint Appendix JA2; * ASHRAE Handbook; *The ACCA Manual J>>	08	Cooling Outdoor Design Temperature Selected (°F)	<< user entry: numeric, x.x degF>>
~~07~~ 09	Heating Outdoor Design Temperature Selected (°F)~~Calculated dwelling unit Sensible Cooling Load (Btu/h)~~	<< user entry: numeric, x.x degF>> ~~<<user entry: integer: xxxxx; or allow selection of value=n/a if value in A06="n/a equipment changeout, like-for-like"~~ >>	~~08~~ 10	Calculated Dwelling Unit Sensible Cooling Load (Btu/h)~~Calculated Dwelling Unit Heating Load (Btu/h)~~	<<user entry: integer: xxxxx; or allow selection of value=n/a if value in A06="n/a equipment changeout, like-for-like">>
11	Calculated Dwelling Unit Heating Load (Btu/h)	<<user entry: numeric: xxxxx (allow n/a entry if "n/a equipment changeout, like-for-like" is selected in A06) >>	~~09~~ 12	Dwelling Unit Number of Bedrooms	<<<<calculated field: integer xx: if CertComplianceType=performance, then use as default the value referenced from CF1R-PRF or allow user to override the default and input a new value constrained to be greater than or equal to the default value from the CF1R-PRF; flag non-default values and report in project status notes field; elseif parent is not CF1R-PRF doc type, then user input the integer value xx>>

	Determination of Mech01 type (this field not visible to user)	<<calculated field: if1 CertComplianceType=performance, then if2 CF1R-PRF Project Scope=one of the following two types: **Addition and/or Alteration **Newly Constructed - Addition Alone then display doc variation MCH-01d; elseif CF1R-PRF Project Scope=Newly Constructed, then display doc variation MECH01a endif2 elseif CertComplianceType=prescriptive additions/alterations, then display doc variation MECH01b, elseif CertComplianceType=prescriptive newly constructed, then display doc variation MECH01c (this field not visible to user) endif1>>			

Notes:
- The outdoor design temperatures for heating shall be ≥99.0% Heating Dry Bulb or the Heating Winter Median of Extremes values.
- The outdoor design temperatures for cooling shall be ≤1.0% Cooling Dry Bulb and Mean Coincident Wet Bulb values.

<<If the project includes an addition, complete fields A13–A16. These fields do not apply to alterations. If CF1R does not include any additions, display the message: "Fields A13–A16 do not apply.">>

13	Maximum Heating Capacity According to Table 150.2-A (Btu/h)	<< user entry: numeric: xxxxx or N/A; if A13 = N/A then the system cannot be ductless and the MCH-23 must be required>>	14	Maximum Cooling Capacity According to Table 150.2-B (Btu/h)	<< user entry: numeric: xxxxx or N/A; if A13 = N/A then the system cannot be ductless and the MCH-23 must be required>>
15	Envelope Leakage Specified in Load Calculation (ACH)	<<User input: DecimalNonnegative, allow from: *Table 150.2-C (only allow if value ≤ the values in Table 150.2-C) *Field Verification and Diagnostic Testing >>	16	Source of Envelope Leakage Specified in Load Calculation	<< user select one from the following values: *Table 150-C; *Field Verification and Diagnostic Testing >>

Note:
- In fields A13 and A14,
  1. If the airflow is verified to be 350 CFM/ton or higher there is no maximum capacity (Section 150.2(a)1Eii) and user should enter N/A.
  2. If the space conditioning system is ductless (Exception 1 to Section 150.2(a)1Eii), user should enter N/A.

MCH-01b - Space Conditioning Systems Ducts and Fans - Prescriptive Alterations

B. Space Conditioning (SC) System Information
<< require one row of data to be entered in this table for each of the quantity of space conditioning systems entered in A04>>

01	02	03	04	05	06	07	08	09	10
SC System ID/Name from CF1R	SC System Description of Area Served	CFA served by this SC System (ft^2):	Is the SC system a ducted system?	Does the work include installing a refrigerant containing component?	Does the work include installing new SC System components?	Does the work include installing more than 25 feet of ducts?	Does the work include installing an “entirely new duct system”?	Does the work include installing an “entirely new SC system”?	Alteration Type
<<reference values from CF1R as default; allow user to override the default and input a new value; flag non-default values and report in project status notes field; a revised CF1R may be required>>	<<reference values from CF1R as default; allow user to override the default and input a new value; flag non-default values and report in project status notes field; a revised CF1R may be required Require each entry to be unique in this dwelling unit i.e. unique within the scope of this instance of the MCH-01>>	<<reference value from CF1R as default; allow user to override the default and input a new value; flag non-default values and report in project status notes field; a revised CF1R may be required >>	<<reference value from CF1R as default; allow user to override the default and input a new value; flag non-default values and report in project status notes field; a revised CF1R may be required >>	<<reference value from CF1R as default; allow user to override the default and input a new value; flag non-default values and report in project status notes field; a revised CF1R may be required >>	<<reference value from CF1R as default; allow user to override the default and input a new value; flag non-default values and report in project status notes field; a revised CF1R may be required >>	<<reference value from CF1R as default; allow user to override the default and input a new value; flag non-default values and report in project status notes field; a revised CF1R may be required >>	<<reference value from CF1R as default; allow user to override the default and input a new value; flag non-default values and report in project status notes field; a revised CF1R may be required >>	<<reference value from CF1R as default; allow user to override the default and input a new value; flag non-default values and report in project status notes field; a revised CF1R may be required If B09 = yes/true AND A06 = n/a like for like, then do not allow user to proceed.>>	<< Calculated field: determine the correct result for "alteration type" for entry in this field by the user responses in B04, B05, B06, B07, B08, B09 and use of Logic Table for Determining Alteration Type and ~~HERS~~ECC Verification Requirements (inserted below this section); constrain user input for fields B04-B09 to allow only the available combinations of responses given in the Logic Table in rows a through t; alteration types are: *Extension of Existing Duct System; *Altered Space Conditioning System; *Entirely New or Complete Replacement Duct System with or without Equipment Changeout; *Entirely New or Complete Replacement Space Conditioning System * No alteration Performed >>

Logic Table for Determining Alteration Type and ~~HERS~~ECC Verification Requirements (this table not shown on the completed document)

	1	2	3	4	5	6	7	8	9
	Is the altered or installed system a ducted system?	Altering or installing a refrigerant containing component?	Installing new components? (packaged unit, or condensing unit, or cooling/heating coil, or air-handling unit, etc)	Installing more than 25 linear feet of new or replacement ducts?	Is the entire duct system accessible for sealing, and is more than 75% of the duct system new or replaced?	Are all of the system's components and ducts new or replaced? (entirely new system)	alteration type	~~HERS~~ECC	notes
a	no	yes	no	no	no	no	Altered space conditioning system	RC	e.g. alteration to refrigerant containing component - mini-split or packaged AC
b	no	yes	yes	no	no	no	Altered space conditioning system	RC	e.g. changeout mini-split system component
c	yes	no	yes	no	no	no	Altered space conditioning system	DctLk	e.g. new ducted hydronic fan coil unit or furnace
d	yes	no	yes	yes	no	no	Altered space conditioning system	DctLk	e.g. new furnace + duct alteration
e	yes	yes	no	no	no	no	Altered space conditioning system	RC	e.g. alteration to a refrigerant containing component - split system
f	yes	yes	yes	no	no	no	Altered space conditioning system	RC + DctLk	e.g. changeout refrigerant containing components
g	yes	yes	yes	yes	no	no	Altered space conditioning system	RC + DctLk	e.g. changeout refrigerant containing component + altered ducts
h	yes	yes	no	yes	no	no	Altered space conditioning system	RC + DctLk	e.g. alteration to refrigerant containing component + altered ducts
i	yes	no	no	yes	yes	no	Entirely new duct system with or without Equipment Changeout	DctLk + FE/AF or Tbl150.0-B,C	e.g. new duct system without equipment changeout
j	yes	no	yes	yes	yes	no	Entirely new duct system with or without Equipment Changeout	DctLk + FE/AF or Tbl150.0-B,C	e.g. new furnace + new duct system
k	yes	yes	no	yes	yes	no	Entirely new duct system with or without Equipment Changeout	RC + DctLk + FE/AF or Tbl150.0-B,C	e.g. alteration to a refrigerant containing component + new duct system
l	yes	yes	yes	yes	yes	no	Entirely new duct system with or without Equipment Changeout	RC + DctLk + FE/AF or Tbl150.0-B,C	e.g. changeout refrigerant containing component + new duct system
m	no	no	yes	no	no	yes	Entirely new space conditioning system	none	e.g. entirely new ductless hydronic heating system (boiler heating only);

										or new wall heater
n	no	yes	yes	no	no	yes	Entirely new space conditioning system	RC	e.g. new mini-split (weigh-in); or new room packaged AC (factory charged)
o	yes	no	yes	yes	yes	yes	Entirely new space conditioning system	DctLk	e.g. new ducted hydronic heating system, or other new ducted heating-only system
p	yes	yes	yes	yes	yes	yes	Entirely new space conditioning system	RC + DctLk + FE/AF or Tbl150.0-B,C	e.g. new split system
q	yes	no	no	yes	no	no	Extension of an existing duct system	DctLk	e.g. altered ducts
r	no	no	no	no	no	no	System is exempt from the alteration requirements	none	no alteration performed
s	yes	no	no	no	no	no	System is exempt from the alteration requirements	none	no alteration performed
t	yes	yes	yes	no	yes	yes	Entirely new space conditioning system	RC + DctLk + FE/AF or Tbl150.0-C,D	e.g. new ducted system that has less than 25 ft of ducts
u	no	no	yes	no	no	no	Altered space conditioning system	none	e.g. new ducted hydronic fan coil unit or new hydronic heating boiler

Nomenclature:
RC = Refrigerant Charge Verification (MCH-25)
DctLk = Duct Leakage Test (MCH-20)
FE/AF or Tbl150.0-B,C - Fan Efficacy and Airflow Rate verification (MCH-22; MCH-23) or alternative compliance: (MCH-28)

C. Space Conditioning (SC) System Alterations Compliance Information

<< require one row of data in this table for each of the SC Systems listed in Section B for which Alteration Type in B10≠ no alteration performed >>

01	02	03	04	05	06	07	08	09	10	10b	11	12	13	14
SC System ID/Name from CF1R	SC System Description of Area Served	Heating System Type	Altered Heating Component	Heating Efficiency Type	Heating Minimum Efficiency Value	Cooling System Type	Altered Cooling Components	Cooling Efficiency Type	Cooling Minimum Efficiency Value SEER/ SEER2	Cooling Minimum Efficiency Value EER/EER2/ CEER	Required Thermostat Type	Number of Indoor Units for this System	Number of Ducted Indoor Units for this System	Central Fan Integrated (CFI) Ventilation System Status
<<reference value from B01>>	<<reference value from B02>>	<< reference value from CF1Ras default; allow user to override the default and pick one from list: *central gas furnace; *central split HP; *central packaged HP *central large packaged HP *ductless mini-split HP; *room HP; *boiler; *hydronic; *combined hydronic; *hydronic+forced air; *combined hydronic+forced air; *hydronic HP, *hydronic HP+forced air; *gas wall furnace; *gas space heater; *electric; *non-air-source heat pump; *Wood Heat; *Small duct high velocity HP; *Ductless VRF HP; *Packaged gas furnace; *multisplit HP-ducted; *multisplit HP-ductless; *multisplit HP-ducted+ductless; *VCHP-ducted; *VCHP-ductless; *VCHP-ducted+ductless; *ducted mini-split HP; *no heating; *SPVHP; *PTHP; if value =no heating, then check: there must be at least one heating system entered in this section in column C04 to comply, else flag noncompliant condition (no heating installed) and do not allow registration to continue. flag non-default values and report in project status notes field; a revised CF1R may be required >>	<< reference value from CF1R as default; allow user to override the default and pick as many as are applicable from list: *gas furnace AHU; *fancoil AHU; *outdoor condensing unit; *indoor coil; *boiler; *TXV or EXV; *compressor; *refrigerant lineset; *no heating component altered; flag non-default values and report in project status notes field; a revised CF1R may be required >>	<<**if** C04 = no heating component altered, **then** value =n/a **elseif** C03 = one of the following system types: *hydronic, *hydronic + forced air, *combined hydronic, *combined hydronic + forced air, *hydronic HP, *hydronic HP + forced air, **then** value=NA; **else** reference value from CF1R as default; allow user to override the default and pick one from list: *AFUE; *HSPF~~;~~ *HSP2; *COP; flag non-default values and report in project status notes field; a revised CF1R may be required >>	<<**if** C04 = no heating component altered, **then** value =n/a **elseif** C03 = one of the following system types: *hydronic, *hydronic + forced air, *combined hydronic, *combined hydronic + forced air, *hydronic HP, *hydronic HP + forced air, **then** value=NA; **else** reference value from CF1R as default; allow user to override the default to enter value: xx.x; note: default minimum value for AFUE=80; and default minimum value for HSPF=8.0; or default minimum value for HSPF2=6.7; flag non-default values and report in project status notes field a revised CF1R may be required>>	<<reference value from CF1R as default; allow user to override the default and pick one from list: *central split AC; *central split HP *central packaged AC ; *central packaged HP *central large packaged AC *central large packaged HP *ductless mini-split AC; *ductless mini-split HP; *gas absorption AC *room AC; *room HP; *hydronic HP, *hydronic HP+forced air *evaporative - direct *evaporative - indirect *evaporative – indirect/direct *evaporatively cooled condenser *Ice Storage AC *non-air-source heat pump; *non-air-cooled air conditioner; *no cooling; *Small duct high velocity HP; *Small duct high velocity AC; *Ductless VRF HP; *Ductless VRF AC; *multisplit AC-ducted; *multisplit AC-ductless; *multisplit AC-ducted+ductless; *multisplit HP-ducted; *multisplit HP-ductless; *multisplit HP-ducted+ductless; *VCHP-ducted; *VCHP-ductless; *VCHP-ducted+ductless; *ducted mini-split AC; *ducted mini-split HP; *SPVAC; *SPVHP; *PTAC; *PTHP; flag non-default values and report in project status notes field; a revised CF1R may be required >>	<< reference value from CF1R as default; allow user to override the default and pick as many as are applicable from list: *outdoor condensing unit, *indoor fancoil AHU, *indoor coil, *TXV or EXV, *Compressor, *refrigerant lineset, *no cooling component altered; flag non-default values and report in project status notes field; a revised CF1R may be required >>	<<reference value from CF1R as default; if C08= no cooling component altered, then value =n/a else allow user to override the default;to enter value: user pick from list: *EER/SEER; *EER2/SEER2; *CEER; flag non-default values and report in project status notes field; a revised CF1R may be required >>	<<reference value from CF1R as default; if C08= no cooling component altered, then value =n/a else allow user to override the default to enter value: xx.x; default minimum value for EER/SEER=14; default minimum values for EER2/SEER2=13.414.3; allow user to overwrite default value, but flag non-default values and report in project status notes field a revised CF1R may be required >>	<<reference value from CF1R as default; If C08 = no cooling component altered, then value = n/a; Else allow user to override the default to ener value: xx.x; default minimum value for EER and EER/SEER = 11.7; default minimum value for EER2 and EER2/SEER2=11.2; default minimum value for CEER = 8.7; allow user to overwrite default value, but flag non-default values and report in project states notes field a revised CF1R may be required >>	<<If alteration type in B10=Extension of Existing Duct System; then display result: "N/A"; else as default display result: "setback" allow user to override the default and pick one from list: *setback; *Occupant Controlled Smart Thermostat (OCST) per JA5; *Energy Management Control System (EMCS)>> >>	<< **if** [C03 or C07] = one of the following system types: *room HP *gas wall furnace; *gas space heater; *electric ; *Wood Heat; *Packaged gas furnace *central packaged AC ; *central packaged HP *central large packaged AC ; *central large packaged HP *room AC; *room HP; *evaporative - direct *evaporative - indirect *evaporative - indirectdirect **then** text value=N/A, elseif C14=CFI System' then integer value=1, else default integer value =1; allow user to overwrite the default to enter one of the following two: 1: an integer value greater than 1, 2: text value=N/A >>	<<**if** B04=no, **then** text value = "n/a"; **else** default integer value =1; allow user to overwrite the default to enter an integer value ≥1 and ≤ the value in C12>>	<<user pick one from list: *CFI System *Not CFI>>

Notes:

D. Installed Heating Equipment Information for Gas Furnace Indoor Unit, or Heat Pump Indoor Unit, or Packaged Unit (Gas Furnace or Heat Pump)

<< **If** all systems listed in Section C have a value in C04= [no heating component altered[, **then** display the section does not apply message;
else for each of the SC systems in Section C for which C04≠[no heating component altered], do the following actions: A, B, C:
A: if C12 >1, then require one row of data in this table for each of the quantity of indoor units specified in C12,
B: if C12=1, then require one row of data in this table,
C: if C12=N/A, then require one row of data in this table>>

01	02	03	04	05	06	07	08	09	10
								Multi-split systems only	
SC System ID/Name from CF1R	SC System Description of Area Served	Heating Efficiency Type	Heating Efficiency Value	Indoor Unit or Packaged Unit Manufacturer	Indoor Unit or Packaged Unit Model Number	Indoor Unit or Packaged Unit Serial Number	SC System Rated Heating Capacity, Output (Btu/h)	Indoor Unit Name or Description of Area Served	Indoor Unit Duct Status
<<reference value from B01>>	<<reference value from B02>>	<<reference value from C05>>	<<if C06 = NA, then report NA; Else user input, numeric, xx.x; check value must be ≥ value in C06, to comply; else flag non-compliant value and do not allow this document to be registered >>	<<user input alphanumeric text string max 50? characters>>	<<user input alphanumeric text string max 50? characters>>	<<user input alphanumeric text string max 50? characters>>	<<user input, numeric, xxxx>>	<<**If** C12=one of the following two values: {N/A}, {1}, **then** value=N/A, **elseif** value in C12>1 then prompt user to input text, 15 characters maximum; do not allow duplicate values in this column on this MCH-01 >>	<<**if** D09=NA, **then** text value=N/A, **else** user pick one of the following three values: 1:[Ductless] 2:[Ducted >10ft length] 3:[Ducted ≤10ft length]>>

Notes:

E. Installed Cooling Equipment Information for Outdoor Condenser or Package Unit (Air Conditioner or Heat Pump):
<<**if** all of the SC Systems listed in Section C have a value in C07=no cooling, **then** display the section does not apply message;

else require one row of data in this table for each of the SC Systems listed in Section C that do not have one of the following two conditions: 1:[a value in C07=no cooling] or 2:[a value in C08 = no cooling component altered]>>

01	02	03	04	04b	05	06	07	08	09
SC System ID/Name from CF1R	SC System Description of Area Served	Cooling Efficiency Type	Cooling Efficiency Value SEER/SEER2	Cooling Efficiency Value EER/EER2/CEER	Condenser or Package Unit Manufacturer	Condenser or Package Unit Model Number	Condenser or Package Unit Serial Number	System Cooling Capacity at Design Conditions (Btu/h)	Condenser Nominal Capacity (tons)
<<reference value from B01>>	<<reference value from B02>>	<<reference value from C09>>	<< if C10 = NA, then report N/A; Else user input, numeric, xx.x; check value must be ≥ value in C10 to comply; else flag non-compliant value and do not allow this document to be registered>>	<< if C10b = N/A, then report N/A; Else user input numeric: xx.x; check value must be ≥ value in C10b to comply; flag non-compliance value and do not allow this document to be registered>>	<<user input alphanumeric text string max 50? characters>>	<<user input alphanumeric text string max 50? characters>>	<<user input alphanumeric text string max 50 characters>>	<<user input, numeric, xxxxxx>>	<<user input, numeric, x.x>>

Notes:

F. Altered Space Conditioning System Duct Information (<75% of duct system is altered; or duct system is not altered)

<<**if** there are no systems in Section B for which BOTH of the following two conditions are true: 1:[B04 = yes] 2:[the Alteration Type in column B10 equals one of the following two {Extension of Existing Duct System}; {Altered Space Conditioning System}], **then** display the "section does not apply" message;

else for each SC System in Section B for which BOTH of the following two conditions are true: 1:[B04 = yes] 2:[the alteration type value in column B10 is equal to one of the following two: {Extension of Existing Duct System}; {Altered Space Conditioning System}], do the following actions A, B, C:

A: require one row of data in this table for each space conditioning system in section D field D02 for which C03=Packaged Gas Furnace.
B: require one row in this table for each space conditioning system in section E field E02 that meets the following two conditions: 1:[value in C07= one of the packaged unit types {central packaged AC}, {central large packaged AC}, {central packaged HP}, {central large packaged HP}], 2:[the same packaged unit is not already listed in section D thus D02≠E02];
C: for systems for which C13≥1, enter one row of data in this table for each of the quantity of ducted indoor units specified in C13 for that system>>

01	02	03	04	05	06	07	08	09	10	11	12
SC System ID/Name from CF1R	SC System Description of Area Served	Indoor Unit Name or Description of Area Served	Was Any New Ducting Installed?	Required New Duct R-Value	Installed New Supply Duct Location	Installed New Supply Duct R-Value	Installed New Return Duct Location	Installed New Return Duct R-Value	Exception from Min R-Value	Can Approved Airflow Protocols be used to test this System?	Indoor Unit Nominal Cooling Capacity (tons)
<<reference value from B01>>	<<reference value from B02>>	<<**if** system type in C03=[Packaged Gas Furnace], **then** value auto filled from B02, **elseif** system type in [C03 or C07] = one of the following four types: 1: central packaged AC ; 2: central packaged HP 3: central large packaged AC ; 4: central large packaged HP **then** value auto filled from B02 elseif value in C12=1, then value autofilled from B02; elseif C12>1, and Section D applies, and D10= one of the following two values: *Ducted >10ft length *Ducted ≤10ft length	<<user pick one of the following two text values: *Both Supply and Return *No *Supply only *Return only>>	<<**if** F04=no, **then** value=n/a, **elseif** B07=no, **then** value = R-6, **elseif** A02= CZ 3, 5-7 **then** value = R-6, **elseif** A02=CZ 1, 2, 4, 8-16 **then** value = R-8>>	<<**if** F04=*no or *Return only, **then** value=n/a; elseif F04=*Supply only or *Both Supply and Return, then require user to pick one from the following list: * conditioned space-entirely, *unconditioned attic, *unconditioned crawl space, *controlled ventilation crawl space *unconditioned garage, *unconditioned basement, *outdoors>>	<< **if** F04=*no or *Return only, then value=n/a; elseif F04=*Supply only or *Both Supply and Return, then require user to pick one value from the following list: *R-0; *R-2.1; *R-4.2, *R-6, *R-8, *R-10, *R-12; **check value**: must be ≥ value in F05 to comply subject to the following exception: **if** F10= *Ducts <R-6 entirely in Conditioned Space, **then** <R-6 complies. flag non-compliant values and do not allow registration to proceed if not in compliance.	<<**if** F04=*no or *Supply only, then value=n/a; elseif F04=*Return only or *Both Supply and Return, then require user to pick one from the following list: * conditioned space-entirely, *unconditioned attic, *unconditioned crawl space, *controlled ventilation crawl space *unconditioned garage, *unconditioned basement, *outdoors>>	<< **if** F04=*no or *Supply only, then value=n/a; Elseif F04=*Return only or *Both Supply and Return, then require user to pick one value from the following list: *R-0; *R-2.1; *R-4.2, *R-6, *R-8, *R-10, *R-12; **check value:** must be ≥ value in F05 to comply subject to the following exception: **if** F10= *Ducts<R-6 entirely in Conditioned Space, **then** <R-6 complies. flag non-compliant values and do not allow registration to proceed if not in compliance.	<< **if** F04=no, then value=n/a, **else** default text value=No Exceptions; allow user to override the default and **select *portions of uninsulated or <R-6 ducts in conditioned space**; **ALSO** **if** value in both F06 and F08= conditioned space-entirely, **then also** allow user to select the following value: **Ducts uninsulated or <R-6 ducts entirely in conditioned space	<<**if** system type in C03 or C07 is one of the following four system types: *multisplit HP-ducted *multisplit HP-ducted+ductless *multisplit AC-ducted *multisplit AC-ducted+ductless, **then** value=no, **elseif** system type in C03 or C07 is one of the following system types: *central split AC; *central split HP *central packaged AC ; *central packaged HP *central large packaged AC *central large packaged HP, **then** value=Yes, **else** user pick one of the following two values from list: **yes **no **check:** **if** value=no, **then** report in project status notes	<<if the following three conditions are **ALL true:** **condition 1:** [C12 > 1], **condition 2:** [system type in C03 or C07=one of the following two values: {VCHP-Ducted}, {VCHP - Ducted+Ductless}] **condition 3:** [one of the following two (a, or b below) are true: **a:**{K03=yes and F11=yes}, **b:**{Responses in B04, B05 , B06, B07, B08, B09 and use of Logic Table for Determining Alteration Type and ~~HERS~~ECC Verification Requirements result in the term "DctLk" appears in the ~~HERS~~ECC column}], **then**

		then use corresponding value autofilled from D09, else user input, text, 15 characters maximum; do not allow duplicate values for indoor unit names in this MCH-01 as listed in F03 and G03>>								field that exemption from mandatory ~~HERS~~ECC verification of system airflow has been claimed. Enforcement agency confirmation is recommended.>>	user input numeric value, x.xx, **else** text value= n/a>>

Notes:

G. Installed New or Complete Replacement Duct System information

<<**if** there are no SC Systems listed in Section B for which B08=yes, **then** display the section does not apply message;
else for each space conditioning system in Section B for which B08=yes, do the following actions A, B, C:
A: require one row of data in this table for each space conditioning system in section D field D02 for which C03=Packaged Gas Furnace.
B: require one row in this table for each space conditioning system in section E field E02 that meets BOTH of the following two conditions: 1:[value in C07= one of the packaged unit types: {central packaged AC}, {central large packaged AC}, {central packaged HP}, {central large packaged HP}], 2:[the same packaged unit is not already listed in section D thus D02≠E02];
C: for systems for which C12≥1,enter one row of data in this table for each of the quantity of ducted indoor units specified in C13 for that system>>

01	02	03	04	05	06	07	08	09	10	11	12	13	14	15
SC System ID/Name from CF1R	SC System Description of Area Served	Indoor Unit Name or Description of Area Served	Total Duct Length	Required New Duct R-Value (Unconditioned Space)	Supply Duct Location	New Supply Duct R-Value	Return Duct Location	New Return Duct R-Value	Exception from Min R-Value	Method of Compliance with Airflow and Fan Efficacy Req's in 150.0(m)13	Number of Air Filter Devices on Indoor Unit	Can Approved Airflow Protocols be used to test this System?	Can Approved Fan Efficacy Protocol be used to test this System?	Indoor Unit Nominal Cooling Capacity (tons)
<<reference value from B01>>	<<auto filled from B02>>	<< **if** system type in C03=[Packaged Gas Furnace], **then** value auto filled from B02, **elseif** system type in [C03 or C07] = one of the following four types: 1: central packaged AC ; 2: central packaged HP 3: central large packaged AC ; 4: central large packaged HP **then** value auto filled from B02 **elseif** value in C12=1, **then** value autofilled from B02; elseif C12>1, and Section D applies, and D10= one of the following two values: *Ducted >10ft length *Ducted ≤10ft length then use corresponding value autofilled from D09, else user input, text, 15 characters maximum; do not allow duplicate values for indoor unit names in this MCH-01 as listed in F03 and G03>>	<<user pick one text value from the following 2: *[>10ft] *[≤10ft]>>	<<**if** A02= CZ 3, 5 - 7 **then** value = R-6.; **elseif** A02=CZ 1, 2, 4, 8 - 16 **then** value = R-8>>	<<User pick one from list: * conditioned space-entirely, *conditioned space -except 12ft, *unconditioned attic, *unconditioned crawl space, *controlled ventilation crawl space *unconditioned garage, *unconditioned basement, *outdoors>>	<<user pick from list: *R-0; *R-2.1; *R-4.2 *R-6, *R-8, *R-10, *R-12; **check value**: must be ≥ value in G05 to comply subject to the following exception: **if** G10= *Ducts <R-6 entirely in Conditioned Space, then <R-6 complies; flag non-compliant values and do not allow registration to proceed if not in compliance>>	<<User pick one from list: *conditioned space-entirely, *conditioned space-except 12ft, *unconditioned attic, *unconditioned crawl space, *controlled ventilation crawl space *unconditioned garage, *unconditioned basement, *outdoors>>	<<user pick from list: *R-0; *R-2.1; *R-4.2 *R-6, *R-8, *R-10, *R-12; **check value:** must be ≥ value in G05 to comply subject to the following exception: **if** G10= *Ducts <R-6 entirely in Conditioned Space, then <R-6 complies flag non-compliant values and do not allow registration to proceed if not in compliance>>	<< Default Value=No Exceptions; allow user to override the default and select *portions of uninsulated or <R-6 ducts in conditioned space; **ALSO** if values in both G06 and G08= conditioned space-entirely then also allow user to select the following value: **Ducts uninsulated or <R-6 ducts entirely in conditioned space	<<if C07=no cooling, then text value = "Exempt - No Cooling"; elseif C07= one of the following three system types: **evaporative - direct, **evaporative - indirect, **evaporative - indirectdirect, then text value = "Exempt - Evaporative System"; elseif G13=no, then text value = "Exempt - Approved Protocols are N/A"; **elseif** B08=yes, AND C14=CFI System, then result = ~~HERS~~ECC Verified Fan Efficacy (W/cfm) and Airflow Rate(cfm/ton); else, user select one from the following two values: **[~~HERS~~ECC Verified Fan Efficacy (W/cfm) and Airflow Rate (cfm/ton)] **[~~HERS~~ECC verified Return Duct Design per Table 150.0-B, C] >>	<<user enter integer value>> note: this value will determine number or rows per indoor unit in next section	<<**if** system type in C03 or C07 is one of the following four system types: *multisplit HP-ducted *multisplit HP-ducted+ductless *multisplit AC-ducted *multisplit AC-ducted+ductless, **then** value=no, **elseif** system type in C03 or C07 is one of the following system types: *central split AC; *central split HP *central packaged AC ; *central packaged HP *central large packaged AC *central large packaged HP, **then** value=Yes, **else** user pick one of the following two values from list: **yes **no **check:** **if** value=no, **then** report in project status notes field that exemption from mandatory ~~HERS~~ECC verification of system airflow has been claimed. Enforcement agency confirmation is recommended.>>	<<**if** system type in C03 or C07 is one of the following four system types: *multisplit HP-ducted *multisplit HP-ducted+ductless *multisplit AC-ducted *multisplit AC-ducted+ductless, **then** value=no, **elseif** system type in C03 or C07 is one of the following system types: *central split AC; *central split HP *central packaged AC ; *central packaged HP *central large packaged AC *central large packaged HP, **then** value=Yes, **else** user pick one of the following two values from list: **yes **no **check**: **if** value=no, **then** report in project status notes field that exemption from mandatory ~~HERS~~ECC verification of system fan efficacy has been claimed. Enforcement agency confirmation is recommended.>>	<<**if** the following three conditions are **ALL true**: **condition 1:** [C12 > 1], **condition 2:** [system type in C03 or C07=one of the following two values: {VCHP-Ducted}, {VCHP - Ducted+Ductless}] **condition 3:** [one of the following two (a or b below) are true: **a:**{K03=yes and G13=yes}, **b:**{Responses in B04, B05 , B06, B07, B08, B09 and use of Logic Table for Determining Alteration Type and ~~HERS~~ECC Verification Requirements result in the term "DctLk" appears in the ~~HERS~~ECC column}],**then** user input numeric value, x.xx, **else** text value= n/a>>

Notes:

H. Installed Air Filter Device Information

Mandatory requirements for air filter devices are specified Section 150.0(m)12. The installer shall place a sticker in or near each filter grille/rack that displays the design airflow rate for that filter grille/rack and the maximum allowed clean filter pressure drop at the design airflow rate. This will inform the occupant of the airflow vs pressure drop performance required for replacement air filters.

<<**If** Section G does not apply, **then** display the section does not apply message; **elseif** there are no duct systems in G03 that have a value in G04 equal to ">10ft", then display the section does not apply message;

else require one row of data (each) for the quantity of air filter devices in G12 for each of the duct systems listed in G03 for which the value in G04=[>10ft];

01	02	03	04	05	06	07	08	09	10	11	12	13
SC System ID/Name from CF1R	SC System Description of Area Served	Indoor Unit Name or Description of Area Served	Air Filter Name or Description of Location	Air Filter Rack Type	Design Airflow Rate for Air Filter Device (cfm)	Air Filter Nominal Depth (inch)	Air Filter Nominal Length (inch)	Air Filter Nominal Width (inch)	Air Filter Calculated Nominal Face Area (inch^2)	Air Filter Required Minimum Face Area (inch^2)	Face Area Compliance	Design Allowable Pressure Drop for Air Filter Device (inch W.C.)
<<reference value from B01>>	<<auto filled from B02>>	<<auto filled from G03	<<user input text, maximum 20 characters>>	<<user select from list: *Filter Grille *Mounted at air handler *Mounted in-line between return grille and air handler >>	<<user enter value numeric; xxxx>>	<<user enter integer value ≥1>>	<<user enter integer value ≥1 >>	<<user enter integer value ≥1 >>	<<calculated numeric value= (H08*H09) >>	<<**if** H07=1, **then** calculated value=(H06 ÷ 150) *144, **else** display text value = "specified by system designer"	<<if value in H11= "specified by system designer", then display text value = "specified by system designer"; elseif H10≥H11, then display text: "complies", else display text:"does not comply">>	<<if value in H07=1, then value = 0.1; else user enter value, numeric, x.xx>>

Notes:

I. Air Filter Device Requirements

<<if section H does not apply, then display the section does not apply message; elseif Section H applies, then display section I.

Mandatory Air Filter Device Requirements can be found in Section 150.0(m)12A-E. Some mandatory requirements may apply in addition to those listed below.

<< Static Text – see the Top>>

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

J. ~~HERS~~ECC Verification Requirements for Duct Systems
<<if both sections F and G do not apply, then display the section does not apply message, else require one row of data in this table for each of the indoor units listed in F03; **also** require one row of data in this table for each of the indoor units listed in G03>>

01	02	03	04	05	06	07	08	09
				MCH-20	MCH-21	MCH-22	MCH-23	MCH-28
SC System Identification or Name	SC System Description of Area Served	Indoor Unit Name or Description of Area Served	Exemption From Duct Leakage Requirements	Duct Leakage Test	Duct Location Verification	AHU Fan Efficacy (W/cfm)	AHU Airflow Rate (cfm/ton)	Return Duct Design - Table 150.0-B or C
<<reference value from B01>>	<<auto filled from B02>>	<<auto filled from F03 or G03 as applicable	<< calculated field: Default text Value= "No Exemptions"; allow user to override the default and pick one of the following three text values from list: * Ducts have previously been sealed, tested, and certified by a ~~HERS~~ECC rater; * Duct system has less than 40 ft of duct; * Duct system is insulated or sealed with asbestos; flag non-default values and report in project status notes field; The enforcement agency may require additional documentation as validation>>	<<Calculated field: if J04 ≠ No Exemptions, then display result = no; elseif J04= No Exemptions, then determine the result for this field by the user responses in B04, B05 , B06, B07, B08, B09 and use of Logic Table for Determining Alteration Type and ~~HERS~~ECC Verification Requirements (inserted below section B); constrain user input for fields B04-B09 to allow only the available combinations of responses given in the Logic Table in rows a through t; If the term **"DctLk"** appears in the ~~HERS~~ECC column, then display text result= "yes" in this field (duct leakage test required); elsif the term **"DctLk"** does not appear in the ~~HERS~~ECC column, then display result= "no" in this field >>	<<if the value in F10 or G10 = *ducts uninsulated or <R-6 ducts entirely in conditioned space, then result = yes; **else** display text result="no">>	<< Calculated field: if G14=no then result=no **elseif** the value in G11= "~~HERS~~ECC Verified Fan Efficacy (W/cfm) and Airflow Rate (cfm/ton)", **then** display text result in this field="yes"; **elseif** all of the following five conditions are true: **B09=yes **C07=no cooling **C14=CFI System, **G13=yes, **G14=yes, then result= yes; else display text result="no">>	<< Calculated field: if the value in F11 or G13=no, then result ="no"; **elseif** the value in G11= "~~HERS~~ECC Verified Fan Efficacy (W/cfm) and Airflow Rate (cfm/ton)", **then** display text result in this field="yes"; **elseif** the value inK03=yes, **AND** the value in J09=no, **then** text result in this field=yes; **elseif** all of the following five conditions are true: **B09=yes **C07=no cooling **C14=CFI System, **G13=yes, **G14=yes, then result= yes; else display text result="no">>	<< Calculated field: if the value in G11= [~~HERS~~ECC verified Return Duct Design per Table 150.0-B, C]; then display text result in this field="yes"; else display text result="no">>
Notes:								

K. ~~HERS~~Field Verification Requirements for Space Conditioning Equipment
<<require one row of data in this table for each of the SC Systems listed in Section C>>

01	02	03
		MCH-25
SC System ID/Name from CF1R	SC System Description of Area Served	Refrigerant Charge
<<auto filled from B01>>	<<auto filled from B02>>	<< Calculated field: **If** [C07 or C03] = one of the following 2 values: *non-airsource heat pump *non-air-cooled air conditioner **then** result = no; **else** determine value by the user responses in B04, B05, B06, B07, B08, B09 and use of "Logic Table for Determining Alteration Type and ~~HERS~~ECC Verification Requirements" (inserted below section B); constrain user input for fields B04-B09 to allow only the available combinations of responses given in the Logic Table in rows a through t; **If** the term "RC" appears in the ~~HERS~~ECC column, **and** A02 =one of the CZ values in the following list: 2, 8, 9, 10, 11, 12, 13, 14, 15, then display text result in this field = yes; else display result = no>>
Notes:		

L. Space Conditioning Systems, Ducts and Fans – Mandatory Requirements and Additional Measures
Additional mandatory requirements from Section 150.0 that are not listed here may be applicable to some systems. These requirements may be applicable to only newly installed equipment or portions of the system that are altered. Existing equipment may be exempt from these requirements.
<< Static Text – see the Top>>

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

M. Test of Defrost Delay Timer Setting (Section 150.0(h)6) The installing contractor shall confirm that a heat pump's installer-adjustable Defrost Delay Timer Setting (if it exists) is set to no less than 90 minutes.		
01	Test Applicability. Select the statement describing test applicability for this project: 1. The test applies because the heat pump utilizes an installer adjustable Defrost Delay Timer Setting to control defrost and there are no exceptions. 2. The test does not apply because the heat pump does not utilize an installer-adjustable Defrost Delay Timer Setting to control defrost. 3. The test does not apply because Exception 1. Dwelling units in Climate Zones 6 and 7 applies. 4. The test does not apply because Exception 2 **.** Dwelling units with a conditioned floor area of 500 square feet or less in Climate Zones 3, 5 through 10, and 15 applies.	<<user selects from list: 1. if test applies and no exceptions 2. If test does not apply because no adjustable Defrost Delay Timer Setting 3. If test does not apply because of Exception 1 4. If test does not apply because of Exception 2 5. N/A>>
<<If M01 = 1. (Applies with no exceptions), then display fields 02-06 for completion Else If M01 = 2. (No adjustable timer setting), then only display the static text "This test in not applicable because there is no installer adjustable Defrost Delay Timer Setting to control defrost" Else If M01 = 3. (Exception 1) Then only display the static text "This test in not applicable because of Exception 1"; Else If M01 = 4. (Exception 2) Then only display the static text "This test in not applicable because of Exception 2"; Else If M01 = 5. N/A Then do not display anything else >>		
02	Recording Configuration of Controls. Specify the mechanism for setting the Defrost Delay Timer Setting (for example, the name of defrost delay timer setting in the thermostat setup, or the location and number of the specific dip switch, jumper, or dial that adjusts Defrost Delay timer).	<<user Input: Text >>
03	Record the heat pump's Maximum Available Defrost Delay Timer Setting (minutes).	<< user Input: User Input: IntegerNonnegative >>
04	Record where you set the Defrost Delay Timer Setting (for example, the numeric timer setting, dip switch position, jumper configuration, or dial setting).	<< user Input: Text >>
05	Record where you set the Defrost Delay Timer Setting, in minutes.	<< user Input: IntegerNonnegative >>
06	Confirming Configuration of Controls. If possible, the Defrost Delay Timer Setting must be 90 minutes or greater. Confirm the Defrost Delay Timer Setting is at least 90 minutes.	<< user selects from list: Yes, or No - Do Not Proceed;>>

N. Test of Supplementary Heating Lockout		
The installing contractor shall confirm the heat pump or thermostat is configured to lock out supplementary heating anytime the outdoor air temperature is above 35°F.		
01	Test Applicability. Select the statement describing test applicability for this project: 1. The test applies because the heating system is a heat pump with supplementary heating and there are no exceptions. 2. The test does not apply because the heating system does not include a heat pump with auxiliary heating. 3. The test does not apply because Exception 1. Heat pump being installed in a single family dwelling unit in Climate Zones 7 or 15. 4. The test does not apply because Exception 2. Heat pump being installed in a single family dwelling unit with a conditioned floor area of 500 square feet or less.	<<user selects from list: 1. if test applies and no exceptions 2. If test does not apply because heating system does not include a heat pump with auxiliary heating 3. If test does not apply because of Exception 1 4. If test does not apply because of Exception 2 5. N/A>>
<<If N01 = 1. (Applies with no exceptions), then display fields 02-06 for completion Else If N01 = 2. (the heating system does not include a heat pump with auxiliary heating), then only display the static text "This test in not applicable because there the heating system does not include a heat pump with auxiliary heating" Else If N01 = 3. (Exception 1) Then only display the static text "This test in not applicable because of Exception 1"; Else If N01 = 4. (Exception 2) Then only display the static text "This test in not applicable because of Exception 2"; Else If N01 = 5. N/A Then do not display anything else >>		
02	If supplementary heating is electric resistance, what is its rated capacity (kW)?	<< user input, decimalnonnegative; else N/A>>
03	If supplementary heating is gas, what is its rated input capacity (kBtu/hr)?	<< user input, numeric, xxxxx; else N/A>>
04	Confirming Configuration of Controls. Specify the mechanism for locking out the supplementary heating (for example, the name of supplementary heating lockout setting in the thermostat setup, or the location and number of the specific dip switch, jumper, or dial that adjusts lockout temperature). If there is no mechanism to lock out supplementary heating above a temperature no greater than 35°F this test fails.	<< user Input: Text >>
05	Record supplementary heating lock-out setting (for example, the numeric thermostat lock-out setpoint, dip switch position, jumper configuration, or dial setting).	<< user input: Text >>
06	At what Outdoor Air Temperature (OAT) are the controls configured to begin locking out supplementary heating? If this number is above 35°F this test fails.	<< user entry: numeric, x.x degF; else if N06 value > 35 degF, Do Not Proceed >>